

Sleep and Memory Report (APA 7 Structured Sections)

Introduction

Sleep is often linked to stronger memory consolidation, and targeted memory reactivation (TMR) is proposed to further support recall. This report summarizes a 2x2 between-participants study with Sleep (Sleep, Wake) and Cue (TMR, Control) conditions. Analyses below are descriptive only.

Study Summary

The dataset contains N = 80 participants. The dependent variable is recall_score (0-40).

Design: Sleep vs Wake and TMR vs Control, with 20 participants in each cell.

Method

Participants completed a word-pair recall task after assignment to one sleep condition and one cue condition. No inferential tests are reported here; all values are direct descriptive summaries from the data.

Results

Overall Descriptive Statistics

Overall recall: M = 25.62, SD = 4.60, Median = 25.19, Range = [14.43, 35.64].

Descriptive Statistics by Condition

Sleep: M = 28.17, SD = 4.07 (n=40)

Wake: M = 23.06, SD = 3.58 (n=40)

TMR: M = 27.19, SD = 4.45 (n=40)

Control: M = 24.04, SD = 4.23 (n=40)

Cell-Level Descriptive Statistics (Sleep x Cue)

Sleep-Control: M = 26.49, SD = 3.58

Sleep-TMR: M = 29.85, SD = 3.92

Wake-Control: M = 21.58, SD = 3.36

Wake-TMR: M = 24.54, SD = 3.22

Interactive Figure

An interactive graph is provided in reports/sleep_memory_interactive_graph.html.

Note

All reported values were computed from the provided dataset without added or fabricated results.